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# Data Science Fundamentals

## Linear Regression

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*Before we get started…*

# Practicalities

# Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

# Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



# Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory excercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

# Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



# How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

# Syllabus

| | | | |
|---|---|---|---|
| Week - 1 | Introduction & Python/JupyterLab Setup | Week - 9 | Easter Holiday |
| Week - 2 | Regression - Linear regression | Week - 10 | Easter Holiday |
| Week - 3 | Regression - Polynomial regression | Week - 11 | Clustering - Hierarchical clustering |
| Week - 4 | Classification - Logistic regression | Week - 12 | Dimensionality reduction: Feature Analysis - PCA |
| Week - 5 | Classification - Decision trees | Week - 13 | Regularization: Lasso regression |
| Week - 6 | Classification - kNN | Week - 14 | Recap: Theory + Excercises |
| Week - 7 | Clustering - kMeans | Week - 15 | Exam Preperation + Q&A |
| Week - 8 | Lifesnaps guidance | | |

(subject to change)

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*Now we get to the fun part*

# Linear Regression

# What is regression

In a regression problem we try to understand the behavior (read: analyse / predict) of a certain (continuous) variable (dependent variable) by studying the influence another variable (independent variable) has on it.

**We want to predict Y based on X**

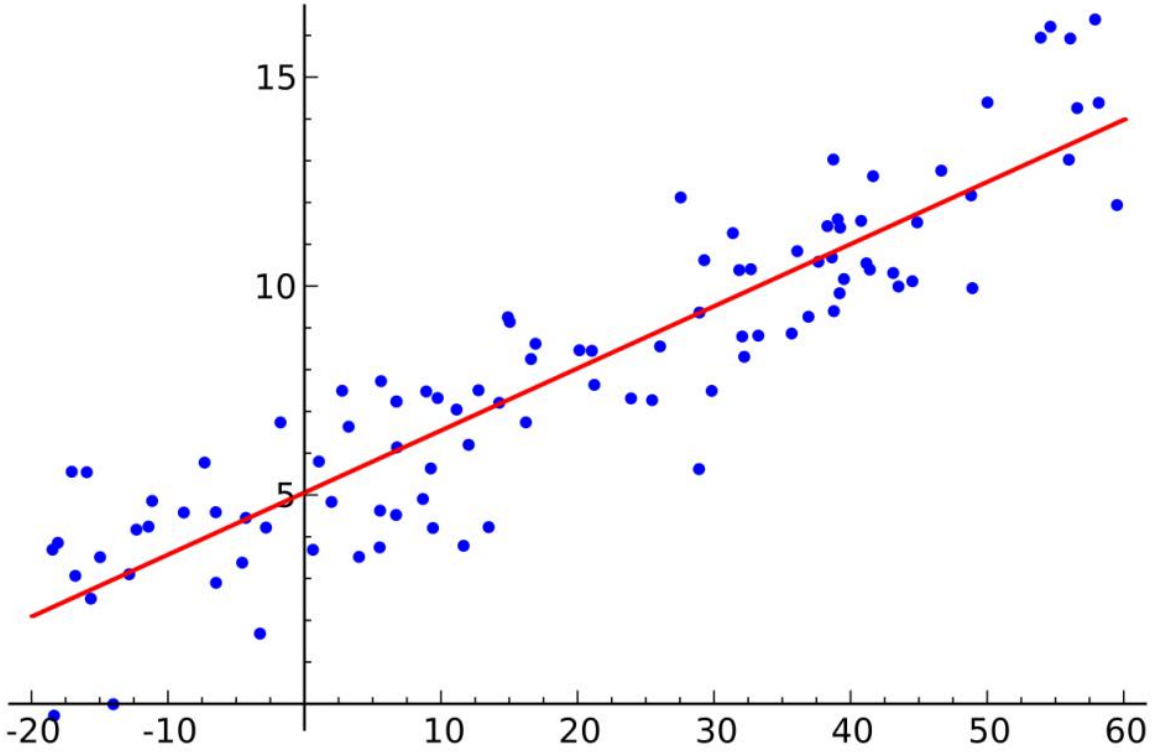


# Examples

- Does “hours studied” affect the variable “exam grade”?
- Does “age” affect “income”?
- Does “muscle mass” affect “time to run a marathon”?
- Does “advertising budget” affect “products sold”?
- …

# Linear regression

- The simplest form of regression
- A linear model → a straight line through the data
- The higher X, the higher (or lower) Y
- "line of best fit"



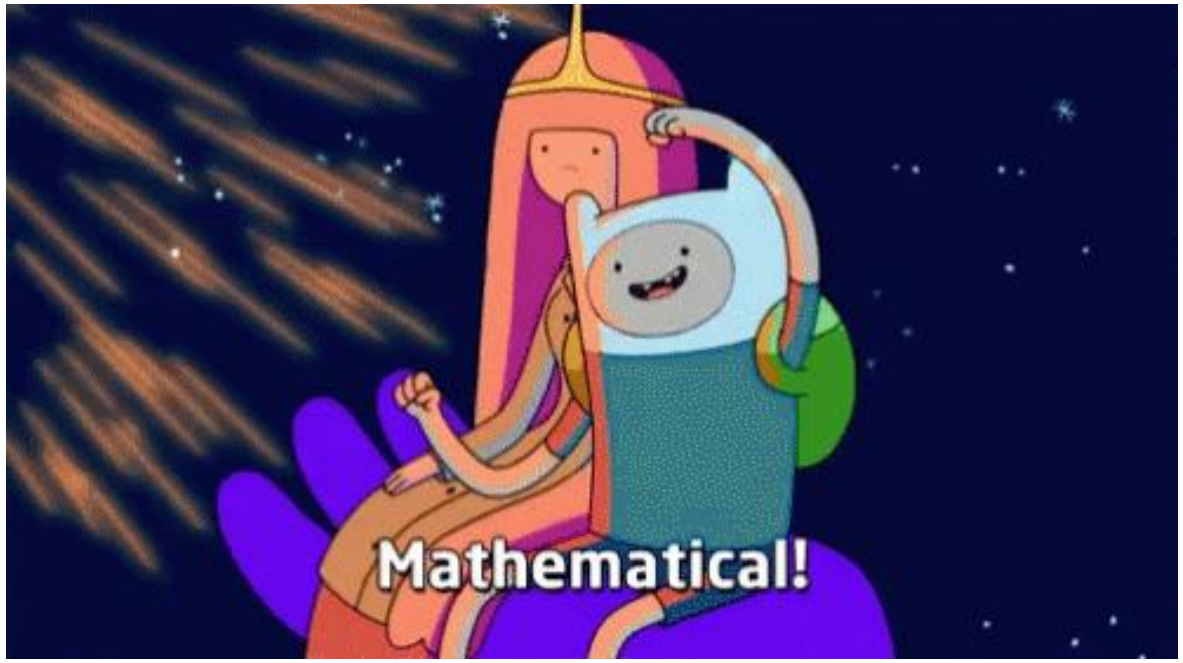
# Linear relation = linear function

- We can describe this behavior as a mathematical function

$$f(x) = ax + b$$

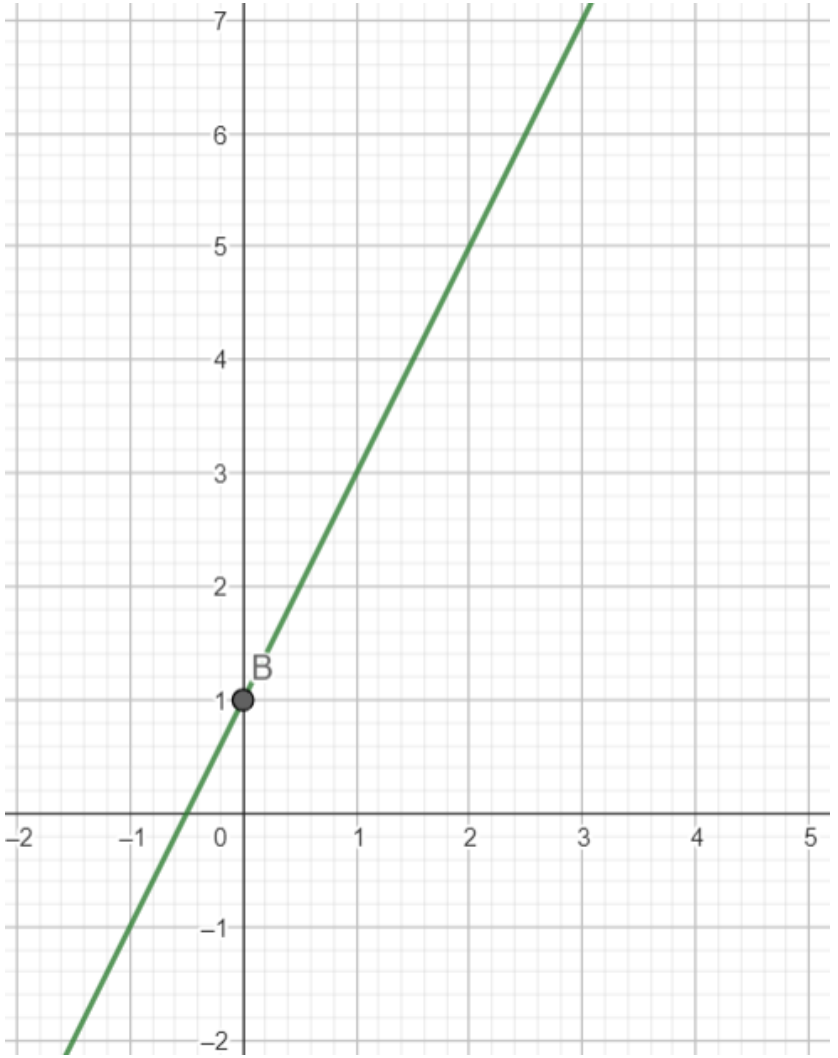
$$y_i = \beta_0 + \beta_1 x$$

- What are Beta 0 and Beta 1?



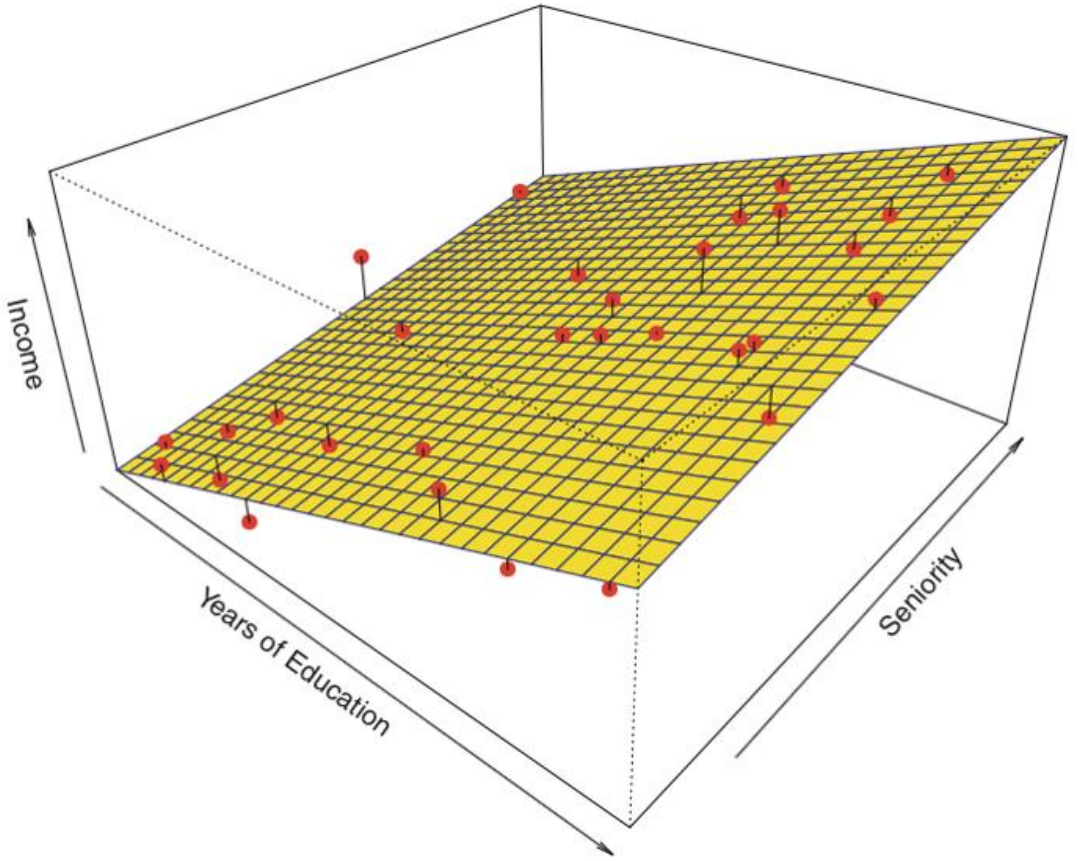
# Linear functions

- Beta 0  is the **intercept**
  - Where the function crosses the X-axis
  - Value of Y when X = 0
    - $y_i = \beta_0 + \beta_1 x$
    - $y_i = \beta_0 + \beta_1 * 0$
    - $y_i = \beta_0$
- Beta 1 is the **slope**
  - Postive Beta 1 → the function grows
  - Negative Beta 1 → the function lowers
  - The increase amount Y with each increase of X



# Multiple linear regression

- Same as linear regression, but with **multiple factors**
- Example: “income” is affected by “seniority” and “years of education”
- “plane of best fit”
- What happens with our function?



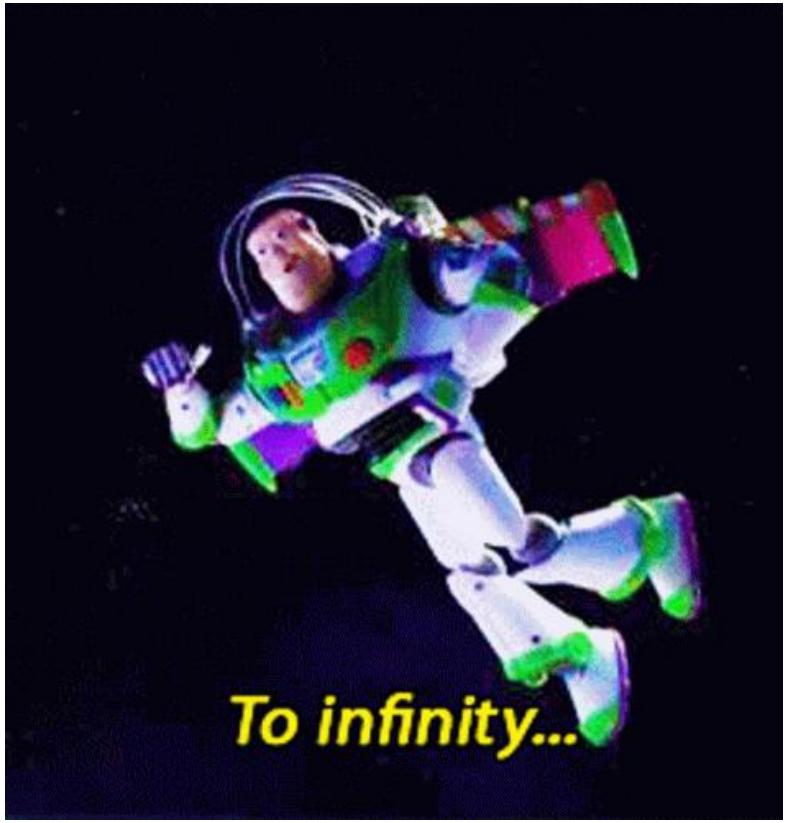

Source: James et al. Introduction to Statistical Learning (Springer 2013)

# Multiple linear regression

- Our intercept remains
- A new “slope” is created for **each parameter**

$$f(x) = ax + bx + cx + dx + \ \ldots + e$$

$$y_i = \beta_0 + \ \beta_1 x + \beta_2 x + \beta_3 x + \beta_4 x + \ldots$$



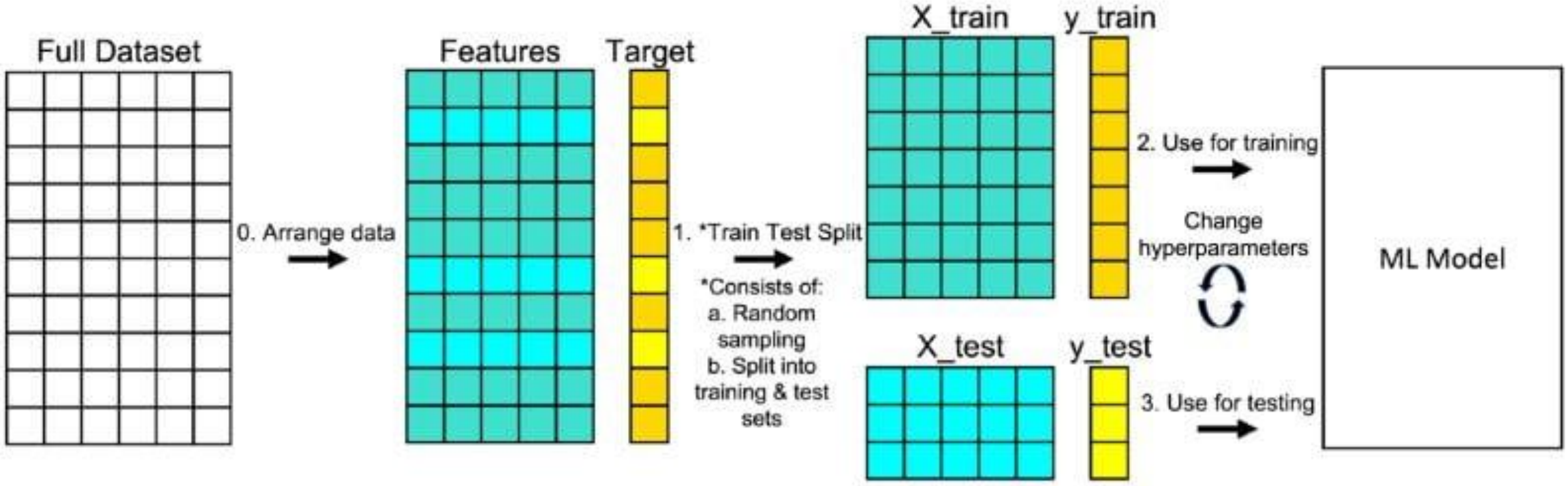
# Training our model

- Do we just train our model on all our data?
- No!
- We split our data




**How does a teacher know you understand the subject if they only ask you questions they've already given you the answer to?**

# Train – Test split




Source: Michael Galarnyk (builtin.com 2022)

# Validating our model

- How do we know whether our model is any good?
- Multiple **metrics**
- Look at the model **itself**:
  - E.g. $R^2$
- **Compare** your model to other models:
  - Mean Absolute Error (MAE)
  - Mean Squared Error (MSE)

# $R^2$

- Only used for **regression**
- Represents the amount of **variation** of the dependant variable (y) explained by the independent variable (x)
- “How much of our data is explained by our model?”
- Always between **0** and **1**
- The **higher**, the better
- Does **not** require **testing data**
- A good indicator of how good a model is **on its own**

# Mean Absolute Error (MAE)

- Represents how accurate a regression model is
- Calculated by the **average** of all the **errors** made in **testing**
- Requires **testing data**
- A good indicator of how good a model when **comparing models**

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | predicted_i - true_i |$$

# Mean Squared Error (MSE)

- Similar to Mean Absolute Error (MAE)
- **Squares** each error
- Metric “punishes” **outliers** → indicate overfitting!
- Often used over MAE

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(predicted_i - true_i)^2$$

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*Let's jump into action*

# Excercises

# First, let's explore this together

- Go to canvas and download the notebook "DSF 02 - Linear Regression Notebook"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



# Now it's up to you!

- Go to canvas and download the notebook "DSF 02 - Linear Regression Exercise"
- Also download the csv file called "Advertising.csv"
- Place them in the same directory
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises

**I don't bite, ask me any questions you have!**

